

Spectrum Sensing System Design

Selected SDR: A selected modular system is aligned with SOSA (Sensor Open Systems Architecture) standards or similar open architectures. This ensures interoperability, allows for future upgrades, and avoids vendor lock-in.

USRP B210:

- Covers the desired frequency range (70 MHz to 6 GHz), including VHF (30-300 MHz) and C-band (4-8 GHz).
- Dual-channel operation allows for simultaneous monitoring or advanced signal processing.
- High-speed USB 3.0 ensures efficient data transfer for real-time applications.
- Price 380 • – 1067 •

Jetson Orin Nano:

- GPU acceleration with 128 CUDA cores is ideal for parallel processing tasks like FFT (Fast Fourier Transform) for spectrum analysis.
- Supports machine learning frameworks (e.g., TensorFlow, PyTorch) for advanced signal classification or anomaly detection.
- Compact and power-efficient, making it suitable for portable or embedded systems.
- price 269 (275 TOPS) •

Antenna System Configurations

- *Channel 1:* VHF/UHF Antenna (70 MHz – 3 GHz)
- *Channel 2:* C-Band Antenna (4 GHz – 6 GHz)

Band	Antenna Type	Model Example	Frequency Range	Gain
VHF/UHF	Discone	Comet DS-150B GHz 156 •	25 MHz – 1.3	2–3 dBi
VHF/UHF	Log-Periodic	Aaronia HyperLOG 7060 GHz 919 •	700 MHz – 6 GHz	5–7 dBi
C-Band	Horn Antenna	Pasternack PE9851	4–8 GHz	10–15 dBi
C-Band	Helical	Custom/DIY	4–6 GHz	12–14 dBi

Table 1: Antenna Selection for Multi-Antenna System

Recommendation: Add wideband log-periodic (HyperLOG 4080, 400 MHz–8 GHz). 476 + (SMA 5mCable) 106 + (SMA to N adapter) 28 •

Supporting Hardware

i) Low Noise Amplifiers

Model	Range	Gain	NF
Nooelec Sawbird+ 85 •	Multi-band	20 dB	<1 dB
Mini-Circuits ZX60-33LN-S+ 159 •	50 MHz–3 GHz	20 dB	3 dB
SparkGap LNA4ALL	28 MHz–2.5 GHz	20 dB	1.5 dB

Note: Sawbird+ units are band-specific (with SAW filters), not truly multiband—select the right variant per band or use wideband LNAs (e.g., Mini-Circuits low-NF parts) plus external filters.

ii) Automatic Antenna Switching System

- RF Switch: Mini-Circuits USB/GPIO-controlled RF switch
 - ZSWA-4-30DR+ (DC–3 GHz, 4-port)
 - Koarmy Model:QM-SW-4S (0.1–6G) 51 •
- Preselection (filters/attenuators) to prevent overload in real-world RF environments
 - FM broadcast notch (88–108 MHz) for near strong broadcast towers.
 - Bandpass tiles for VHF/UHF, 700–960 MHz, 1.7–2.7 GHz, 3–4 GHz, 4–6 GHz.
- Stable timing/clocking for coherence, TDOA, and repeatable measurements
MIMO sync cable For dual-channel phase coherence • 200 •
GPSDO or an external 10 MHz + PPS reference: improves frequency accuracy, stability, and enables multi-node TDOA geolocation and clean cross-site data fusion.
- Digitally controlled step attenuator (e.g., 0–31 dB) placed before the SDR to prevent ADC overload in high-signal environments.

iii) Outdoor/Rooftop Monitoring Outdoor installations require additional considerations for weather resistance and safety:

- *Non-penetrating roof mounts* while providing secure mounting
- *Mast mounting (1.25”–2” diameter)* – Standard mast sizes for antenna installation
- *Weatherproof antenna enclosures* protecting sensitive RF components from environmental exposure
- *Grounding and lightning protection*
- *Quality low-loss cables (LMR-400)* – Minimize signal degradation over longer cable runs

Recommendation:

- **LNAs:** Excellent (Nooelec Sawbird+, ZX60; NF <3 dB)
- **RF Switch:** ZSWA-4-30DR+ (Jetson GPIO control)
- **Outdoor:** LMR-400 2.83 per foot • + lightning protection

Testing & Calibration

- *Antenna Performance Verification*

Proper system characterization requires several key measurements:

VSWR Measurement: Use NanoVNA or similar vector network analyzer to verify antenna impedance matching

Gain Pattern: Compare received signal levels against known reference signals to validate antenna gain

Frequency Response: Perform frequency sweep tests with USRP to characterize system response across bands

Noise Floor: Establish baseline noise measurements for sensitivity analysis

- *Software Tools*

The following software tools are recommended for system testing and operation:

GNU Radio: Spectrum analyzer flowgraph development and signal processing

GQRX: Real-time spectrum visualization and monitoring

URH (Universal Radio Hacker): Advanced signal analysis and protocol decoding

Compute and pipeline

1. **ML Pipeline:** PyTorch anomaly detection. Use cuFFT/cuSignal with pinned memory and zero-copy buffers where possible.
2. Prefer 8-bit IQ for streaming/storage unless dynamic range needs dictate 12/16-bit.
3. Ensure UHD decimation in the FPGA and keep host sample rates modest (e.g., 2–10 Msps per channel) for sustained real-time operation.
4. **UI:** Flask web dashboard with gr-specest waterfalls

Upgrade Cost Matrix

item	Benefit	Cost \$
USRP B210		1,067
Jetson Orin	ML/radar processing	269
Sync Cable		200
comet Ds 150		156
Aaronia Hypewrlog 7060		919
SMA+SMA to NAdap		140
Nooelec Sawbird+		85
Koarmy Model:QM-SW-4S		51
LMR-400 30 foots		100
Total		~ \$3,000
RFSoc	FPGA/DAC integration	+5,000

Table 2: Scalability Options

Similar projects: several SDR systems and projects that target spectrum sensing in the VHF to C-band range, as this spectrum is widely used for telecommunications, broadcasting, and radar applications.

- i) *DARPA Spectrum Collaboration Challenge (SC2):* Focused on developing cognitive radio systems for dynamic spectrum access to monitor and analyze frequencies for detecting primary users and optimizing spectrum usage.

- ii) *Cognitive Radio Testbeds*: Research institutions like the University of California, Berkeley, and Virginia Tech have developed SDR-based testbeds for spectrum sensing.
- iii) *Commercial Solutions*: Companies like ThinkRF and Per Vices offer SDR-based spectrum monitoring solutions that cover VHF to C-band.
- iv) *Open-Source Projects*:
 - GitHub: GNU Radio
 - OpenAirInterface (OAI): An open-source platform for 4G/5G experimentation, which includes spectrum sensing modules. GitHub: OpenAirInterface
 - SigMF (Signal Metadata Format): A standard for storing and sharing SDR data, often used in spectrum monitoring projects.
 - Enhanced Real-Time Processing
 Jetson Nano's GPU for FFT computations with libraries like CuFFT.
 Minimize data transfer bottlenecks between USRP B210 and Jetson Nano by optimizing USB 3.0 throughput.
 - User Interface Development:
 Develop a web-based UI using Flask or Django (Python) for remote access to spectrum data. Alternatively, use additional resources for a desktop application with real-time spectrum visualization:
 gr-specest: GNU Radio spectral estimation toolkit
 LTE Cell Scanner: Frequency scanning implementations
 Pothos SDR: Alternative flow-based SDR framework